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Time : 3.00 hrs

MATHEMATICS

Max Marks : 90

PART – I

(i) All questions are compulsory. **20 x 1 = 20**

(ii) Choose the most appropriate answer from the given four alternatives and write the option code and the corresponding answer.

1. If $A^T A^{-1}$ is symmetric, then A^2 = a) A^{-1} b) $(A^T)^2$ c) A^T d) $(A^{-1})^2$

2. If $A = \begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix}$ and $A(\text{adj } A) = \begin{bmatrix} k & 0 \\ 0 & k \end{bmatrix}$, then k = a) 9 b) $\sin\theta$ c) $\cos\theta$ d) 1

3. The augmented matrix of a system of linear equations is $\begin{bmatrix} 1 & 2 & 7 & 3 \\ 0 & 1 & 4 & 6 \\ 0 & 0 & \lambda-7 & \mu+5 \end{bmatrix}$. The system has infinitely many solutions if a) $\lambda = 7, \mu \neq -5$ b) $\lambda = -7, \mu = 5$ c) $\lambda \neq 7, \mu \neq -5$ d) $\lambda = 7, \mu = -5$

4. The area of the triangle formed by the complex numbers z, iz, and z + iz in the Argand's diagram is
a) $\frac{1}{2}|z|^2$ b) $|z|^2$ c) $\frac{3}{2}|z|^2$ d) $2|z|^2$

5. If $|z_1| = 1$, $|z_2| = 2$, $|z_3| = 3$ and $|9z_1z_2 + 4z_1z_3 + z_2z_3| = 12$, then the value of $|z_1 + z_2 + z_3|$ is a) 1 b) 2 c) 3 d) 4

6. If $z = x + iy$ is a complex number such that $|z + 2| = |z - 2|$, then the locus of z is
a) real axis b) imaginary axis c) ellipse d) circle

7. If $|z + 3 - 4i| \leq 5$, then the greatest value of $|z|$ is a) 5 b) 8 c) 10 d) 12

8. The number of real numbers in $[0, 2\pi]$ satisfying $\sin^4 x - 2\sin^2 x + 1$ is a) 2 b) 4 c) 1 d) ∞

9. The polynomial $x^3 + 2x + 3$ has a) one negative and two imaginary zeros b) one positive and two imaginary zeros c) three real zeros d) no zeros

10. $\tan^{-1}\left(\frac{1}{4}\right) + \tan^{-1}\left(\frac{2}{9}\right)$ is equal to a) $\frac{1}{2}\cos^{-1}\left(\frac{3}{5}\right)$ b) $\frac{1}{2}\sin^{-1}\left(\frac{3}{5}\right)$ c) $\frac{1}{2}\tan^{-1}\left(\frac{3}{5}\right)$ d) $\tan^{-1}\left(\frac{1}{2}\right)$

11. If $\cot^{-1}(\sqrt{\sin\alpha}) + \tan^{-1}(\sqrt{\sin\alpha}) = u$, then cos2u is equal to a) $\tan^2\alpha$ b) 0 c) –1 d) $\tan 2\alpha$

12. If $\sin^{-1}\frac{x}{5} + \text{cosec}^{-1}\frac{5}{4} = \frac{\pi}{2}$, then the value of x is a) 4 b) 5 c) 2 d) 3

13. $\sin^{-1}\frac{4}{5} - \cos^{-1}\frac{12}{13} + \sec^{-1}\frac{5}{4} - \text{cosec}^{-1}\frac{13}{12}$ is equal to a) 2π b) π c) 0 d) $\tan^{-1}\frac{12}{65}$

14. The radius of the circle $3x^2 + by^2 + 4bx - 6by + b^2 = 0$ is a) 1 b) 3 c) $\sqrt{10}$ d) $\sqrt{11}$

15. If P(x, y) be any point on $16x^2 + 25y^2 = 400$ with foci $F_1(3, 0)$ and $F_2(-3, 0)$ then $PF_1 + PF_2$ is a) 8 b) 6 c) 10 d) 12

16. If x + y = k is a normal to the parabola $y^2 = 12x$, then the value of k is a) 3 b) –1 c) 1 d) 9

17. The centre of the circle inscribed in a square formed by the lines $x^2 - 12x + 20 = 0$ and $y^2 - 16y + 63 = 0$ is
a) (6, 8) b) (5, 8) c) (6, 7) d) (5, 7)

18. If $\vec{a}$ and $\vec{b}$ are unit vectors such that $[\vec{a}, \vec{b}, \vec{a} \times \vec{b}] = \frac{1}{4}$ then the angle between $\vec{a}$, and $\vec{b}$ is
a) $\frac{\pi}{6}$ b) $\frac{\pi}{4}$ c) $\frac{\pi}{3}$ d) $\frac{\pi}{2}$

19. The angle between the lines $\frac{x-2}{3} = \frac{y+1}{-2}, z = 2$ and $\frac{x-1}{1} = \frac{2y+3}{3} = \frac{z+5}{2}$ is a) $\frac{\pi}{6}$ b) $\frac{\pi}{4}$ c) $\frac{\pi}{3}$ d) $\frac{\pi}{2}$

20. Find the distance from the origin to the lane 2x + y + 2z – 6 = 0 a) 0 b) 1 c) 2 d) 3

PART - II

Answer any seven questions. Question No.30 is compulsory. **7 x 2 = 14**

21. If adj $A = \begin{bmatrix} -1 & 2 & 2 \\ 1 & 1 & 2 \\ 2 & 2 & 1 \end{bmatrix}$, find A^{-1}.

22. Find the rank of the following matrix by row reduction method. $\begin{bmatrix} 1 & 1 & 1 & 3 \\ 2 & -1 & 3 & 4 \\ 5 & -1 & 7 & 11 \end{bmatrix}$

23. If $|z| = 3$, show that $7 \leq |z + 6 - 8i| \leq 13$

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24. Find the principal argument Arg z, when $z = \frac{-2}{1+i\sqrt{3}}$
25. Find the sum of squares of roots of the equation $2x^4 - 8x^3 + 6x^2 - 3 = 0$
26. Show that the equation $x^9 - 5x^5 + 4x^4 + 2x^2 + 1 = 0$ has atleast 6 imaginary solutions.
27. Find the principal value of $\cos^{-1}\left(\frac{\sqrt{3}}{2}\right)$
28. Obtain the equation of the circle for which (3, 4) and (2, –7) are the ends of a diameter.
29. Determine whether the three vectors $2\hat{i}+3\hat{j}+\hat{k}$, $\hat{i}-2\hat{j}+2\hat{k}$ and $3\hat{i}+\hat{j}+3\hat{k}$ are coplanar.
30. Find the equations of tangent to the parabola $x^2 + 2x - 4y + 4 = 0$ at (0, 1)

PART - III

Answer any seven questions. Questions No.40 is compulsory. **7 x 3 = 21**

31. Solve, by Cramer's rule, the system of equations $5x - 2y + 16 = 0$, $x + 3y - 7 = 0$.
32. Show that $(2+i\sqrt{3})^{10} + (2-i\sqrt{3})^{10}$ is real.
33. If $z = x + iy$ is a complex number such that $\left|\frac{z-4i}{z+4i}\right| = 1$ show that the locus of z is real axis.
34. Solve the equation $x^3 - 3x^2 - 33x + 35 = 0$
35. Find the value of $\sin^{-1}\left(\sin\frac{5\pi}{9}\cos\frac{\pi}{9} + \cos\frac{5\pi}{9}\sin\frac{\pi}{9}\right)$
36. Solve $\tan^{-1}\left(\frac{1-x}{1+x}\right) = \frac{1}{2}\tan^{-1} x$ for $x > 0$.
37. Find the centre and radius of the circle $x^2 + y^2 - x + 2y - 3 = 0$
38. Find the equation of the ellipse with foci $(\pm 2, 0)$, vertices $(\pm 3, 0)$
39. Forces of magnitudes $5\sqrt{2}$ and $10\sqrt{2}$ units acting in the directions $3\hat{i}+4\hat{j}+5\hat{k}$ and $10\hat{i}+6\hat{j}-8\hat{k}$, respectively, act on a particle which is displaced from the point with position vector $4\hat{i}-3\hat{j}-2\hat{k}$ to the point with position vector $6\hat{i}+\hat{j}-3\hat{k}$. Find the work done by the forces.
40. Find the points where the straight line passes through (2, 5, –4) and (6, 1, 4) cuts the xz plane.

PART - IV

Answer all the questions. **7 x 5 = 25**

41. a) Solve the following system of linear equations, by Gaussian elimination method.
$4x + 3y + 6z = 25$, $x + 5y + 7z = 13$, $2x + 9y + z = 1$. **(OR)**

41. b) If $\tan^{-1} x + \tan^{-1} y + \tan^{-1} z = \pi$, show that $x + y + z = xyz$.

42. a) If $2\cos\alpha = x + \frac{1}{x}$ and $2\cos\beta = y + \frac{1}{y}$, show that (i) $\frac{x^m}{y^n} - \frac{y^n}{x^m} = 2i\sin(m\alpha - n\beta)$ (ii) $x^m y^n + \frac{1}{x^m y^n} = 2\cos(m\alpha + n\beta)$ **(OR)**

42. b) Solve the equation $6x^4 - 5x^3 - 38x^2 - 5x + 6 = 0$ if it is known that $\frac{1}{3}$ is a solution.

43. a) If $z = x + iy$ and $\arg\left(\frac{z-1}{z+1}\right) = \frac{\pi}{2}$, show that $x^2 + y^2 = 1$. **(OR)**

43. b) The parabolic communication antenna has a focus at 2m distance from the vertex of the antenna. Find the width of the antenna 3m from the vertex.

44. a) Solve the following systems of linear equations by Cramer's rule.
$\frac{3}{x} - \frac{4}{y} - \frac{2}{z} - 1 = 0$, $\frac{1}{x} + \frac{2}{y} + \frac{1}{z} - 2 = 0$, $\frac{2}{x} - \frac{5}{y} - \frac{4}{z} + 1 = 0$ **(OR)**

44. b) If $\cos^{-1}x + \cos^{-1}y + \cos^{-1} z = \pi$ and $0 < x, y, z < 1$, show that $x^2 + y^2 + z^2 + 2xyz = 1$

45. a) Find the centre, foci and eccentricity of the hyperbola $11x^2 - 25y^2 - 44x + 50y - 256 = 0$ **(OR)**

45. b) Find the non-parametric form of vector equation, and Cartesian equations of the plane.
$\vec{r} = (6\hat{i}-\hat{j}+\hat{k}) + s(-\hat{i}+2\hat{j}+\hat{k}) + t(-5\hat{i}-4\hat{j}-5\hat{k})$

46. a) Using vector method, prove that $\cos(\alpha-\beta) = \cos\alpha\cos\beta + \sin\alpha\sin\beta$ **(OR)**

46. b) Solve : $(x - 5)(x - 7)(x + 6)(x + 4) = 504$

47. a) Show that the Cartesian equation of the plane passing through the points (2, 1, 3), (4, 2, 1) and perpendicular to plane $2x - y + z - 4 = 0$ is $x + 6y + 4z - 20 = 0$ **(OR)**

47. b) Solve the equation $z^3 - 64i = 0$, where $z \in C$.

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